

Project Name	AI Driven Testing
Online team meeting	https://fau.zoom.us/j/69553438847?pwd=VHGm0oOZ2KvPbbhrGsmX83d4rAuuaT.1
Production system (if any)	...
Test system (if any)	...
GitHub repository	https://github.com/amosproj/amos2025ss04-ai-driven-testing
GitHub feature board	https://github.com/orgs/amosproj/projects/82/views/2
GitHub imp-squared backlog	https://github.com/orgs/amosproj/projects/86/views/1
Planning Poker Link	https://pokerplanning.org/room/05e2b4ba-b452-4c29-8ba0-e9ae23005ce0
Team T-shirt (white)	https://www.shirtinator.ch/s/gyMoSd27QOSIIB0wUYQ7XA
Team T-shirt (black)	https://www.shirtinator.ch/s/-GuNOvW5Q2qjHFDZrYILpA
black link again since it didnt work for some	https://www.shirtinator.ch/t-shirts/gestalten/t-shirt-bedrucken#/load/share/f86b8d3a-f5b9-436a-a31c-50d9ad820ba4
Additional materials	...
Team mailing list	oss-amos-proj4@lists.fau.de

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Alsultan	Moaiad	Moaiadsu	m.alhmadhalsultan@campus.tu-berlin.de

#	Meeting Day	Product Owner		Software Developer	Release Manager	Scrum Master	First-Level-Support :)	Comment
		Review	Planning					
1	2025-04-16	both	both	all other	none	Felix	Felix	
2	2025-04-23	both	Max	all other	Max	Felix	Felix	
3	2025-04-30	Max	Alex	all other	Alex	Felix	Felix	
4	2025-05-07	Alex	Max	all other	Jonas	Felix	Felix	Build process review
5	2025-05-14	Max	Alex	all other	Lennard	Felix	Felix	
6	2025-05-21	Alex	Max	all other	Lisabeth	Felix	Felix	
7	2025-05-28	Max	Alex	all other	Tessa	Felix	Felix	Mid-term due
8	2025-06-04	Alex	Max	all other	Max	Felix	Felix	
9	2025-06-11	Max	Alex	all other	Alexander	Felix	Felix	
10	2025-06-18	Alex	Max	all other	Biranavan	Felix	Felix	
11	2025-06-25	Max	Alex	all other	Aditi Vishwas	Felix	Felix	
12	2025-07-02	Alex	Max	all other	Moaiad	Felix	Felix	
13	2025-07-09	Max	Alex	all other	Jonas	Felix	Felix	
14	2025-07-16			all other	Lennard	Felix	Felix	Demo day!
Product owners, software developers, and Scrum Master are set and ideally don't change over time; the critical part is the Release Manager role you need to define here								

Goals	Interpersonal relationship objectives: "To foster an atmosphere of mutual respect and learning, creating a team-spirit"
	Main goal is to create a satisfying project that makes Us, the IP(Industry Partner) and open-source community happy
Meeting norms	Documentation in english and meeting language in German
	Start on time
	Be Polite
	Purpose: Clearly define the goal of the meeting. Preparation: Ensure all participants are well-informed and ready to contribute. Participation: Encourage active engagement from all attendees. Process: Organize the meeting structure and agenda effectively. Progress: Monitor the meeting's progress and ensure it stays on track.
Working norms	Leave the code better than you found it (The Boy Scout Rule)
	Keep it Simple, Stupid (KISS)
Coordination norms	Felix will do the retros + keeping track of meetings
	Alex and Max will take care of the assignments (Backlog Items)
Communication norms	We check Discord at least every day
Consideration norms	Communicate Conflict in Team-Meeting. Help from Felix or Jovana to solve conflicts.
	Respect, active listening, tolerance
Cont. improvement norms	Experimentation spirit and feedback culture
	Burn-Down charts as process tracking
Rewards	Appreciation and praise
Sanctions	Wear the ducky tie
	Post a cute/funny pic of a pet (or similar)
Signatures	
Scrum Master	Felix Lang
Product owner	Maximilian Krug
Product owner	Alexander Lorenz
Software developer	Mohammad Moaiad Alhamdh Alsultan (Moaiad Alsultan)
Software developer	Jonas Brüggemann
Software developer	Lennard Clicqué
Software developer	Birnavan Parameswaran (Biri)
Software developer	Lisabeth Hasse
Software developer	Tessa Heidkamp
Software developer	Aditi Vishwas Takale

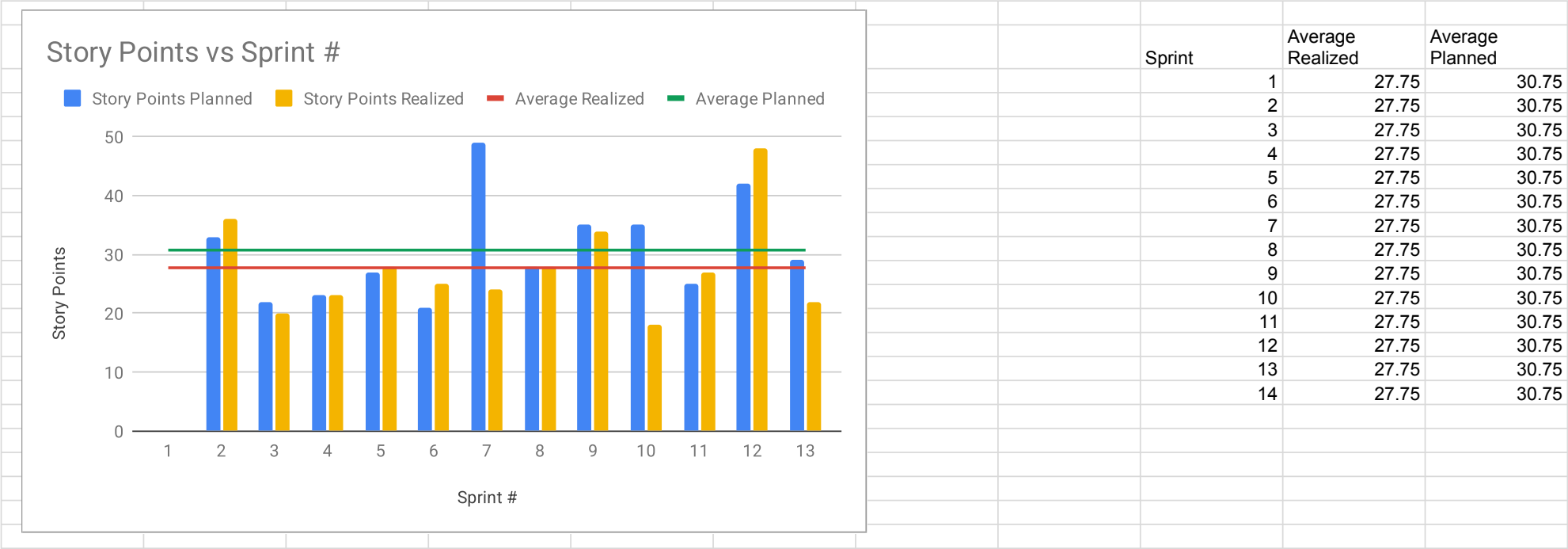
Product Vision	Project Mission
Product Vision We are transforming the landscape of software testing through intelligent, AI-powered automation. Our vision is a future where testing is no longer a bottleneck but a seamless, continuous, and reliable part of the development process. By empowering developers and testers with AI tools that autonomously generate and maintain test cases, we free them from generating the same test cases over and over to instead focus on creative problem-solving and strategic innovation. Our solution supports a future of scalable, secure, and locally controlled quality assurance, driving the next evolution in software development. We envision the role of the programmer to become a curator and innovator of AI generated Code.	Product Mission Deliver a minimum viable product (MVP) of an AI-powered test generation assistant that interacts effectively with real-world software projects. This includes laying the empirical and research groundwork for the idea. From this research we will conclude a MVP, that can analyze simple Python and C++ programs and produce corresponding test cases. Operated through a user-friendly chat interface. The MVP will showcase initial integration with widely used open-source testing frameworks such as Robot Framework, proving the viability of AI-assisted test automation in practical, everyday development environments.

Term	Definition
LLM	Large language model
LLM Input	User Input into the LLM
LLM Output	Response of the LLM to the Input
LLM System Prompt	System message sent (this is different from the LLM Input)
Temperature	gradient how "creative" the LLM behaves
Interface Script/ LLM factory	Backbone script of the project to start, end and use the LLM containers
Container	Docker container, (not kaniko os something else)
IP	Industry Partner
Sprint	agile (Amos) sprint with a duration of 2 weeks
CI	GitHub workflows as Continuous Integration assistant
Wiki	Documentation in the Github repos wiki (not Wikipedia)
Laptop	Personal device to run the code (nothing fancy standard device) TODO add specs
API	Application Programming Interface
UI	User Interface
UX	User Experience
Database	Organized collection of data
Cloud Computing	On-demand delivery of computing services over the internet
Machine Learning	Field of AI that enables systems to learn from data
AI	Artificial Intelligence
Data Analytics	Process of examining data to draw conclusions
Big Data	Extremely large data sets analyzed to reveal patterns
Cybersecurity	Protection of internet-connected systems from digital attacks
DevOps	Software development and IT operations practices
Agile	Iterative approach to software development
Scrum	Framework for agile project management
NLP	Natural Language Processing
Frontend	Part of a website or application that users interact with
Backend	Server-side of a website or application
Full Stack	Developer proficient in both frontend and backend
Version Control	System that records changes to a file or set of files over time
Git	Distributed version control system in our case Github
Repository	Storage location for code, documents, or other assets
Compiler	Program that translates source code into executable code
Debugging	Process of finding and fixing errors in code
Algorithm	Set of rules or instructions to solve a problem
Framework	Reusable, standardized set of tools or libraries for software development
Library	Collection of pre-written code that can be used to perform common tasks

Term	Definition
Microservices / Module	Architectural style that structures an application as a collection of services
Serverless Computing	Cloud execution model where the provider dynamically manages server allocation
Ollama	Tool for managing the LLMs github.com/ollama
Virtual Machine	Emulation of a computer system

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Sprint #	Story Points Planned	Story Points Realized
1		
2	33	36
3	22	20
4	23	23
5	27	28
6	21	25
7	49	24
8	28	28
9	35	34
10	35	18
11	25	27
12	42	48
13	29	22
	PLEASE CREATE THE VELOCITY CHART ON A NEW TAB USING THE DATA FROM THIS TAB	

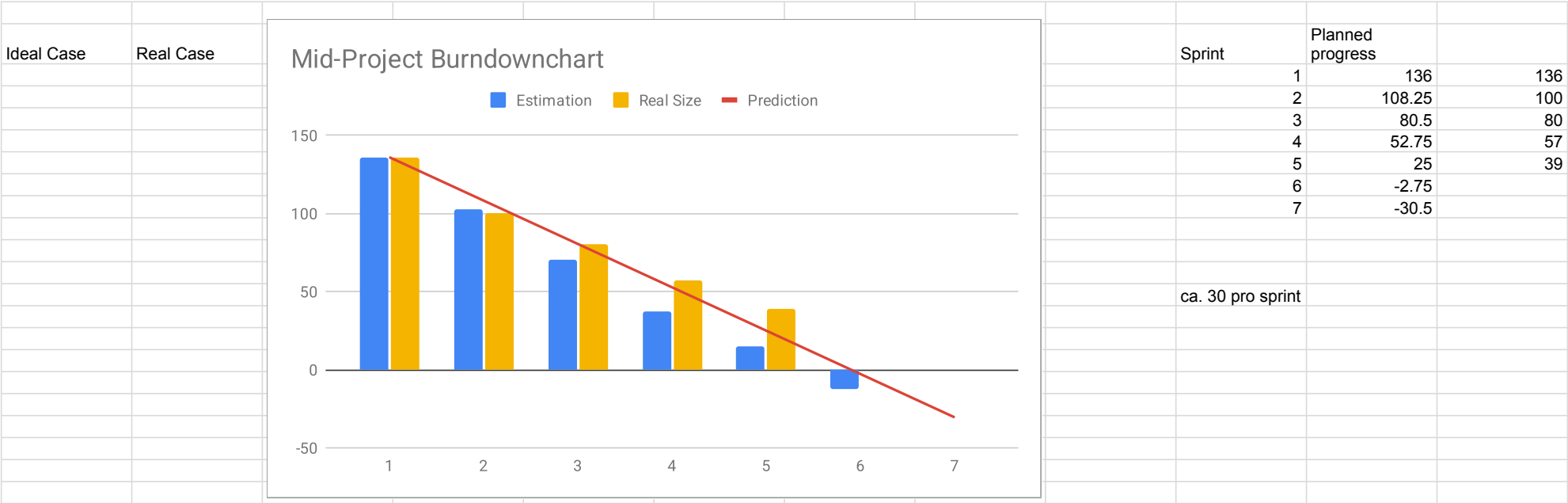


Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			136	136	132	
Sprints						
1			0	136	0	132
2			33	103	36	132
3			33	70	20	96
4			22	37	23	76
5			27	15	28	53
6			21	-12	25	25
7						0
8						0
9						0
10						0
11						0
12						0
13						0
14						0
15						0
Finish						0
Features						
1						
		(none since we didnt meet with the IP yet)				
2						
		Write .py script to interact with the LLM 3	3		3	
		Reaserch LLM 5	5		5	
		Research Deepcoder/Phi4-Mini	5		5	
		Reaserch LLM 4	5		5	
		Reaserch LLM 2	5		8	

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
		Create initial Architecture Documentation	2		2	
		Research LLM 1	5		3	
		Write .py script to interact with the LLM 1	3		5	
3						
		Continue to Maintain the Architecture-Documents	3		3	
		Write .py script to interact with the LLM 2	3		3	
		Research Code Complexity	5		5	
		Create Python test cases	3		1	
		Start Bill of Materials	1		1	
		Continue to Maintain Bill of Materials	1		1	
		Write .py script to interact with the LLM 5	3		3	
		Write .py script to interact with the LLM 4	3		3	
4						
		Continue to Maintain Bill of Materials	1		1	
		Unify the python interface	3		3	
		initialize README	2		1	
		Research AI-Model Benchmark	3		3	
		Build the first Benchmarking tests	3		3	
		make CI-Pipeline	5		8	
		Continue to Maintain the Architecture-Documents	3		2	
		Set up Branchmanagement	3		2	
5						
		Add Modul Interface	5		5	
		Research Ollama	5		3	
		Test docker performance	3		5	
		Initialize API	5		5	
		Build process video	3		3	
		Initialize Frontend	5		3	
		Continue to Maintain Bill of Materials	1		1	
		Make metrics.py into its own modul			3	
6						
		Write .py script to start docker containers	3		3	
		Show POs how the Project works	3		2	
		Research: Spell checker	3		3	
		Onboard small LLM	1		1	
		Frontend Make User enter prompt and source code	5		8	
		Module: Analyze context size	3		5	
		Connect API to Frontend	3		3	

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Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total		estimation ca. 30 per sprint	243	243		
Sprints						
7			49	243	24	243
8			28	194	28	219
9			35	166	34	191
10			35	131	18	157
11			25	96	27	139
12			42	71	48	112
13			29	29	22	64
Features						
7						
		Module. Evaluate LLM performance	3			
		Extract Code from Input/Output	3		2	
		Research/Implement How to include whole project	5		13	
		Module: Code Complexity MCC	5			
		Set up Docker compose	5		5	
		Unify build process	5			
		Bill of Material	1		1	
		Architecture Document	3		3	
		Module CCC	5			
		Initialize user, (technical) design, and build/deploy documentation	3			
		Refactoring: unify payload to json	5			
		Module: CMT Code Complexity	3			
		Set prerequisites for ModuleI	3			
8						
		Fix: keep models	2		2	
		Refactoring: unify payload to json	5		5	
		Bug-Fix: Update Extract Code from Input/Output #128	1		1	
		Module CCC	5		5	
		Module. Evaluate LLM performance	3		5	
		Unify build process	5		3	
		Onboard and Test new LLM	2		2	

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
		Module: Code Complexity MCC	5		5	
9						
		Onboard and Test new LLM	2		2	
		Module: Add configurable outputs	3		3	
		Research: Befits of chaining methods	3		3	
		Test current models for consitency	3		2	
		Module: LLM testing of Code understandability, make LLM describe the code, let the same LLM (with no history) build the code from the description -> see if code passes the Unit tests.	3		3	
		Research: How to measure code coverage with tools	3		3	
		Frontend: make modules togglable in UI	5			
		Frontend: Diff viewer between original and tested source	5		5	
		Module: external research	5		8	
		Initialize user, (technical) design, and build/deploy documentation	3		5	
10						
		Onboard and Test new LLM	3		2	
		Enable Output to Reinput for LLM	5			
		Module: Check output for code coverage of the tests	5			
		Clean up output code	5			
		Research: can we run the models in the CI	3		3	
		Module: Show dependency hierarchy of code by logic flow	3		3	
		Module: execute Tests	5			
		Research: How to train LLM	3		5	
		Research: How to finetune LLM	3		5	
11						
		Research and Run first project on HPC	3		5	
		Module: execute Tests	5		5	

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
		Acquire Dataset: Unit-Tests	3		3	
		Acquire Dataset: Python-Tests	3		3	
		Use metrics to compare 1b, 3b, 7b and maybe 14b models	8		8	
		Set prerequisites for Module1	3		3	
12						
		Clean up output code	5		5	
		Onboard and Test new LLM	2		2	
		Module: Detect redundant/duplicate test cases	3		3	
		Enable Output to Reinput for LLM	5		5	
		Frontend: Upload .py files through the UI	3		3	
		Homework: Prepare Demo and create 3 Usecases	2		2	
		Homework: Create Demo Day slide	3		5	
		Homework: Create Demo Day video	5		8	
		Refactor: Readme files	2		2	
		Refactor: clean repo	5		3	
		Frontend: Display Outputs of Selected Modules	3		5	
		Onboard and Test new LLM	2		3	
		Acquire English-Dataset for LLM	2		2	
13						
		Clean the wiki	3		3	
		Homework: Overwork the Bill of Materials	3		3	
		Homework: Overwork the Architectuer Document	3		5	
		<i>{user design build}-documentation.pdf</i>	3		3	
		Eveluate the performance of the models with and without the modules	3		3	
		Check Docker: streamline build process	3			
		Module: Check output for code coverage of the tests	5			
		Demoday Prep: Cheat sheet	3		5	
		(Module:) Output Exporter	3			

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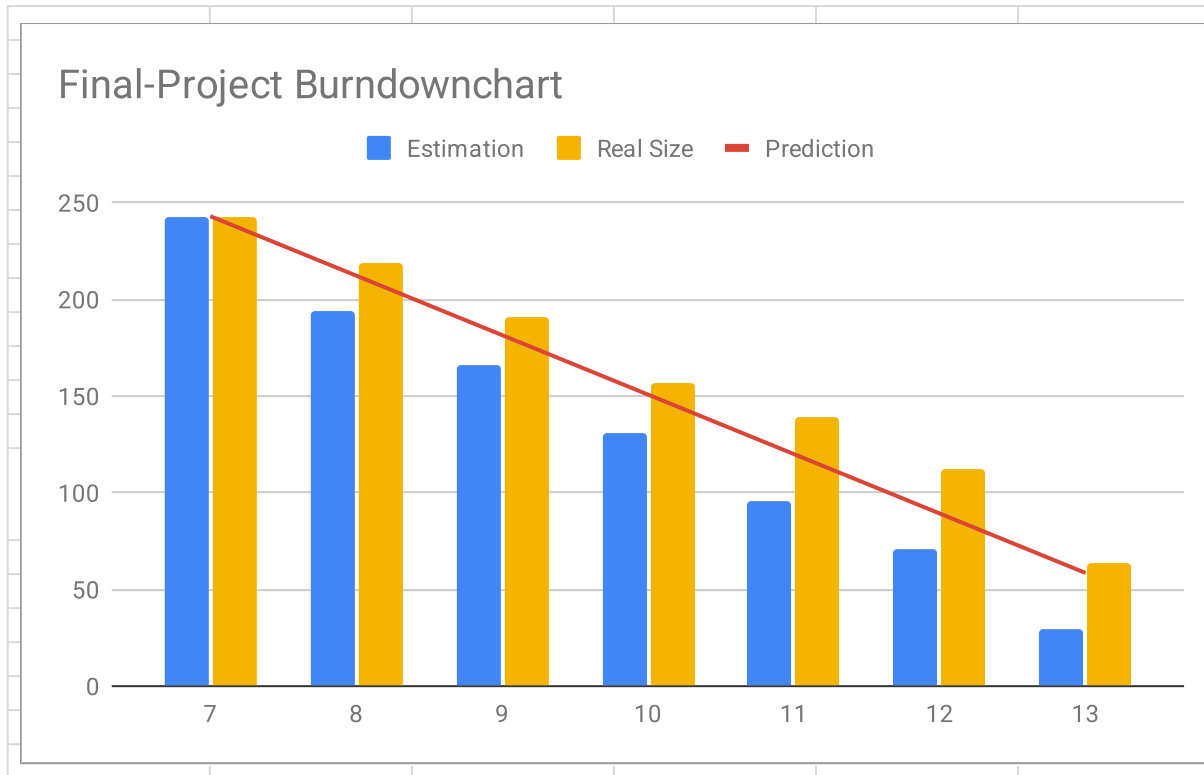
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Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining



	Sprint	Prediction
	7	243
	8	212.25
	9	181.5
	10	150.75
	11	120
	12	89.25
	13	58.5
	Note: only the current sprints are accurate	It is accurate for the final release

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Type	Link / reference
Wiki	https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki
Architecture	https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Architecture
Build and Deployment Guide	https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Build-and-Deployment-Guide
Top-Level Readme:	https://github.com/amosproj/amos2025ss04-ai-driven-testing
Overview for models	https://github.com/amosproj/amos2025ss04-ai-driven-testing/wiki/Comparison-of-1b,-3b,-7b-and-14b-models
Documentation (e.g Architecture)	https://github.com/amosproj/amos2025ss04-ai-driven-testing/tree/development/Documentation
Deliverables (e.g Homework)	https://github.com/amosproj/amos2025ss04-ai-driven-testing/tree/development/Deliverables

	Name	Version	License	Comment	
Main Tools					
	Python	3.13.2	Python Software Foundation License Version 2	For running the main script	
	Docker	28.1.1	Apache License, Version 2.0	Ollama enables running LLMs easily	
	Ollama	0.6.8	MIT License		
	conda	25.5.0	BSD-3 License	environment management	
	conda-forge	25.5.0	BSD-3 License	Distribution channel	
Used LLMs (via Ollama)					
	StarCoder2	starcoder2:3b	Apache License, Version 2.0	Running StarCoder2 3B via Ollama	
	Qwen2.5-coder	qwen2.5-coder:3b-instruct-q8_0	Qwen Research License (3B Model only), Apache License 2.0 (others)	Running Qwen2.5-Coder via Ollama	
	Mistral	mistral:7b-instruct-v0.3-q3_K_M	Apache License, Version 2.0	Running Mistral via Ollama	
	OpenHermes 2.5	openhermes:v2.5	Apache License, Version 2.0	Running OpenHermes via Ollama	
	Phi4-Mini	phi4-mini:3.8b-q4_K_M	MIT License	Running Phi4-Mini via Ollama	
	Phi4-reasoning	phi4-reasoning:14b	MIT License	Running Phi4-reasoning via Ollama	
	TinyLlama	tinylama:1.1b	Apache License, Version 2.0	Running TinyLlama via Ollama	
	Qwen3	qwen3:4b-q4_K_M	Apache License, Version 2.0	Running Qwen3 via Ollama	
	Smollm2	smollm2:360m	Apache License, Version 2.0	Running Smollm2 via Ollama	
Backend					
	pytest	8.3.1	MIT License	used for unit-tests	
	docker	7.1.0	Apache License, Version 2.0		
	requests	2.32.3	Apache License, Version 2.0		
	tqdm	4.67.1	MIT License		
	numpy	1.22.0	BSD License		
	pandas	1.3.0	BSD License		
	fastapi	0.115.1	MIT License		
	uvicorn[standard]	0.34.0	BSD License		
	transformers	4.52.2	Apache License, Version 2.0		
	langchain	0.3.25	MIT License		
	langchain_community	0.3.24	MIT License		
	langchain-chroma	0.2.4	MIT License		
	langchain_text_splitters	0.3.8	MIT License		
	langchain_ollama	0.3.24	MIT License		
	keybert	0.9.0	MIT License		
	langchain	0.9.0	MIT License		
	beautifulsoup4	4.13.4	MIT License		
	py2cfg	1.0.5	Apache License, Version 2.0		
	mccabe	0.7.0	MIT License		
	pydantic	2.11.7	MIT License		
	staticfg	0.9.5	Apache License, Version 2.0		
	Black	25.1.0	MIT License	Used for Code formatting	
	flake8	7.0.0	MIT License	Used for Linting	
Frontend					
	@emotion/react	11.14.0	MIT License		

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Last Name	First Name	Value					
Brüggemann	Jonas	5		5.00	OK		
Clicqué	Lennard	5					
Hasse	Lisabeth	5					
Heidkamp	Tessa	5					
Parameswaran	Birnavan	5		0	No size		
Takale	Aditi Vishwas	5		1	Trivial size		
				2	Small size		
				3	Medium size		
				5	Large size		
				8	Very large size		
				13	Too large (size)		
How to play planning poker							
1. Everyone type their number into their value field, don't hit return yet							
2. Someone, perhaps a product owner, count down 3.. 2.. 1..							
3. Then, everyone hit return to submit their value							